MIL-C-50330B(AR)
30 October 1979
SUPERSEDING
MIL-C-50330A(MU)
15 April 1971

MILITARY SPECIFICATION

CONVERTER DC/DC: 10549021

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the dc/dc converter used in Power Supply PP-4812/VPS-2, a part of Radar Set AN/VPS-2. The dc/dc converter is also used in Subassembly PP6324/TPM-22, part of Radar Test Set AN/TPM-22. The dc/dc converter provides the necessary voltages for functional operation of the Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-46357	Computer, Gun Direction: M18 Manufacturing Processes and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1285

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND,(ARRADCOM)

F10549021	Dc/Dc Converter

PACKAGING DATA SHEET

10549021	Packaging of Dc/Dc Converter

(Copies of specifications, standards, drawings, and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. The dc/dc converter shall be manufactured in accordance with Drawing F10549021 and drawings pertaining thereto.

3.2 General specifications. Where applicable, the dc/dc converter shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

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3.2.2 MIL-C-46357.

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. The dc/dc converter shall be mounted on the vibration table in normal service position. The dc/dc converter shall meet the requirements of 3.4 and 3.5 after being subjected to continuous vibration for 2 hours along each of three mutually perpendicular axes within the frequency range of 10 to 40 hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency at each frequency range shall not exceed 1 Hz per second. Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. The dc/dc converter shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 + 1 milliseconds (ms). The G value of each impulse shall be within $\pm$ 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5.5 ms.

3.3.3 Altitude.

3.3.3.1 Operating. The dc/dc converter shall meet the requirements of 3.4 and 3.5 during and after exposure to the barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. The dc/dc converter shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to the barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The dc/dc converter shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +155°F. The adjusted output

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voltages shall be allowed to vary ± 5% at temperature extremes. The nominal adjusted maximum currents shall be allowed to increase 5% of the specified maximum values at temperature extremes.

3.3.4.2 Nonoperating (storage). The dc/dc converter shall not suffer permanent damage when subjected to not less that 4 hours at each temperature extreme of -65°F and +160°F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (nonoperating). The dc/dc converter shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity of 94 ± 4 percent for a period of not less than 10 days.

3.4 Electrical.

3.4.1 Continuity. Each conductor shall be electrically continuous as specified on the schematic diagram of the dc/dc converter, F10549002.

3.4.2 Resistance. Resistance between terminals shall be as listed in Table I.

TABLE I. RESISTANCE

TERMINALS		RESISTANCE (IN OHMS)
FROM	TO	
E1	E2 (Positive lead)	Less than 10 ohms
E1	E3	Open
E2	E3	Open
J4-1	J4-2	Short
J4-1	J4-9	Short
J4-3	J4-10	Short
J4-3	J4-11	Short
J4-4	J4-5	* 0.65 to 3 ohms
J4-4	J4-1	Open
J4-14	J4-15	Short
J4-7	J4-8	Short
J4-12	J4-13	* 0.65 to 3 ohms
J4-12	J4-14	Open
J3-1	J3-2	* 0.65 to 3 ohms
J3-1	J3-7	Open

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TABLE I. RESISTANCE - CONTINUED

TERMINALS		RESISTANCE (IN OHMS)
FROM	TO	
J3-3	J3-4	* 0.65 to 3 ohms
J3-3	J3-9	Open
J2-1	J2-2	* 0.65 to 3 ohms
J2-1	J2-7	Open
J2-3	J2-4	* 0.65 to 3 ohms
J2-3	J2-9	Open
J1-1	J1-2	* 0.65 to 3 ohms
J1-1	J1-7	Open
J1-3	J1-4	* 0.65 to 3 ohms
J1-3	J1-9	Open

* NOTE: The difference in value among any of these eight readings shall not exceed 0.5 ohms.

3.5 Performance. The dc/dc converter shall produce the outputs given in Table II when loads are applied as specified in Table II and input voltage to the dc/dc converter is 22 to 30 volts direct current (Vdc).

3.6 First Article Inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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TABLE II. OUTPUT VOLTAGE

TERMINALS FROM (+)	TERMINALS TO (-)	RESISTIVE LOAD OR OUTPUT CURRENT	OUTPUT VOLTAGE	AC RIPPLE	WAVEFORM
J4-1,2,9	J4-3,10,11	10.0 ± 0.3A	35 to 53 Vdc	Spikes equal or less than average dc output voltage	N/A
J4-7,8	J4-14,15	3.50 ± 0.15A	9 to 14 Vdc		N/A
J4-7,8	J4-6	0.1A ± 10%	18 to 28 Vdc		N/A
J1-6	J1-7	1.00 ± 0.15A	19 to 29 Vdc		N/A
J1-8	J1-9	1.00 ± 0.15A	19 to 29 Vdc		N/A
J2-6	J2-7	1.20 ± 0.15A	26 to 38 Vdc		N/A
J2-8	J2-9	1.20 ± 0.15A	26 to 38 Vdc		N/A
J3-6	J3-7	0.75 ± 0.15A	36 to 53 Vdc		N/A
J3-8	J3-9	0.75 ± 0.15A	36 to 53 Vdc		N/A
J4-4	J4-5	2k ohms	60-90 V p-p	N/A	Square Wave
J4-12	J4-13	2k ohms	60-90 V p-p	N/A	
J1-1	J1-2	2k ohms	60-90 V p-p	N/A	
J1-3	J1-4	2k ohms	60-90 V p-p	N/A	
J2-1	J2-2	2k ohms	60-90 V p-p	N/A	
J2-3	J2-4	2k ohms	60-90 V p-p	N/A	
J3-1	J3-2	2k ohms	60-90 V p-p	N/A	
J3-3	J3-4	2k ohms	60-90 V p-p	N/A	
J4-4	J4-5	2k ohms	60-90 V p-p	N/A	
J4-12	J4-13	2k ohms	60-90 V p-p	N/A	

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inpections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) submission. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three dc/dc converters manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provision (SQAP) listed in the Technical Data Package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 and 4.4 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.3.2.3 and 4.4 shall consistute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests. The requirements and tests in Table III shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at standard ambient temperatures (60°F to 90°F).

TABLE III. CLASSIFICATION OF DEFECTS
(100 percent inspection)

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR:			
101.	Electrical	3.4	4.6.2
102.	Performance	3.5	4.6.3
MINOR: NONE.			

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4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. One dc/dc converter shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table IV. Three dc/dc converters shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table V.

TABLE IV. SPECIAL SAMPLING

NO.	CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301.	Fabrication	3.1	Applicable drawings - Visual
302.	General specifications	3.2	MIL-F-13926 & MIL-C-46357 - Visual

TABLE V. SPECIAL SAMPLING, ENVIRONMENTAL

NO.	CONTROL TEST	REQUIREMENT	TEST PROCEDURE
303.	Vibration	3.3.1	4.6.1.1
304.	Shock	3.3.2	4.6.1.2
305.	Altitude	3.3.3	4.6.1.3
306.	Temperature	3.3.4	4.6.1.4
307.	Humidity (nonoperating)	3.3.5	4.6.1.5

4.4.1 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

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4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Test equipment. In conjunction with 4.5.2.2, the standard test equipment listed in Table VI shall be utilized in performing the applicable tests of 4.6.

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TABLE VI. TEST EQUIPMENT LIST

ITEM NUMBER		DESCRIPTION
1.	Voltmeter	Dc, 0-50 V, 1% accuracy, 20,000 ohms/volt or greater.
2.	Ammeter	Dc, 0-15 A, 1% accuracy.
3.	Oscilloscope	Vertical passband: dc to 1 MHz. Horizontal deflection: 5 us/cm. Input sensitivity: 20 mv/cm. Input impedance: 1 megohm.
4.	Resistive load, fixed	2K ± 10% ohms, 2 W.
5.	Resistive load, variable	0 to 50 ohms, 1000 W.
6.	Resistive load, variable	0 to 300 ohms, 100 W.
7.	Ohmmeter	To measure less than 3 ohms or infinity.
8.	Power supply	Dc, 22-30 V, 20 A, regulated to ± 0.5%.

4.6 Test methods and procedures.

4.6.1 Environmental. Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810. After performing the requirements of paragraphs 4.6.2 and 4.6.3 at room ambient conditions, nominal settings shall be established with no further adjustments at environmental extremes. Voltage and current specifications shall be allowed to degrade per 3.3.4.1.

4.6.1.1 Vibration. This test shall be performed as specified in Procedure VIII, Method 514 of MIL-STD-810 as modified by paragraph 3.3.1 herein. After vibration, the dc/dc converter shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.1.

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4.6.1.2 Shock. This test shall be performed as specified in Procedure I, Method 516 of MIL-STD-810 except that shock pulses shall be 11.0 ms and 15 G. After the shock impulses, the dc/dc converter shall be tested as specified in 4.6.2 and 4.6.3 for compliance with 3.3.2.

4.6.1.3 Altitude (operating and nonoperating). This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that the operating altitude shall be limited to 10,000 feet (20.6 in. Hg). Operation of Step 4 shall be the performance tests specified in 4.6.3 herein. Nonoperating test of 40,000 feet (5.5 in. Hg) shall also be performed.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and nonoperating). This test shall be performed as specified in Procedure I, Method 501 of MIL-STD-810. The temperature of Step 3 shall be maintained for not less than 4 hours. The temperature of Step 4 shall be +155°F. In Step 5, the temperature shall be maintained for 4 hours and the dc/dc converter shall be tested in accordance with 4.6.2 and 4.6.3 during this time.

4.6.1.4.2 Low (operating and nonoperating). This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -65°F maintained for not less that 4 hours. The temperature of Step 4 shall be -40°F. In Steps 5 and 7, the dc/dc converter shall be tested in accordance with 4.6.2 and 4.6.3.

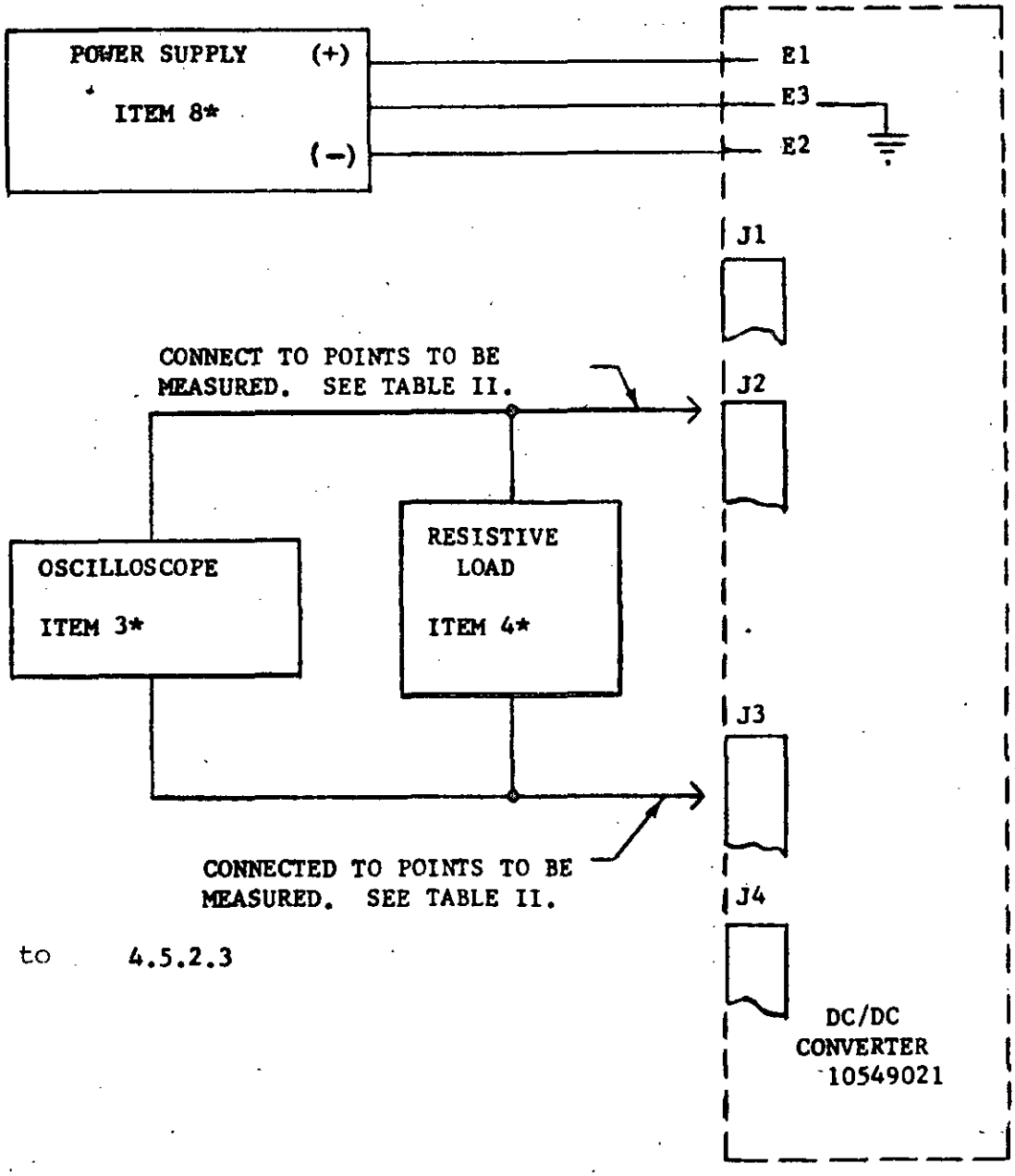
4.6.1.5 Humidity (nonoperating). This test shall be performed in accordance with Procedure II, Method 507 of MIL-STD-810. The dc/dc converter shall be tested in accordance with 4.6.3.

4.6.2 Electrical.

4.6.2.1 Continuity. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.1.

4.6.2.2 Resistance. Resistance shall be measured with an ohmmeter, Item 7, for compliance with 3.4.2.

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*Refer to 4.5.2.3

FIGURE 1. DC/DC CONVERTER TEST SETUP.

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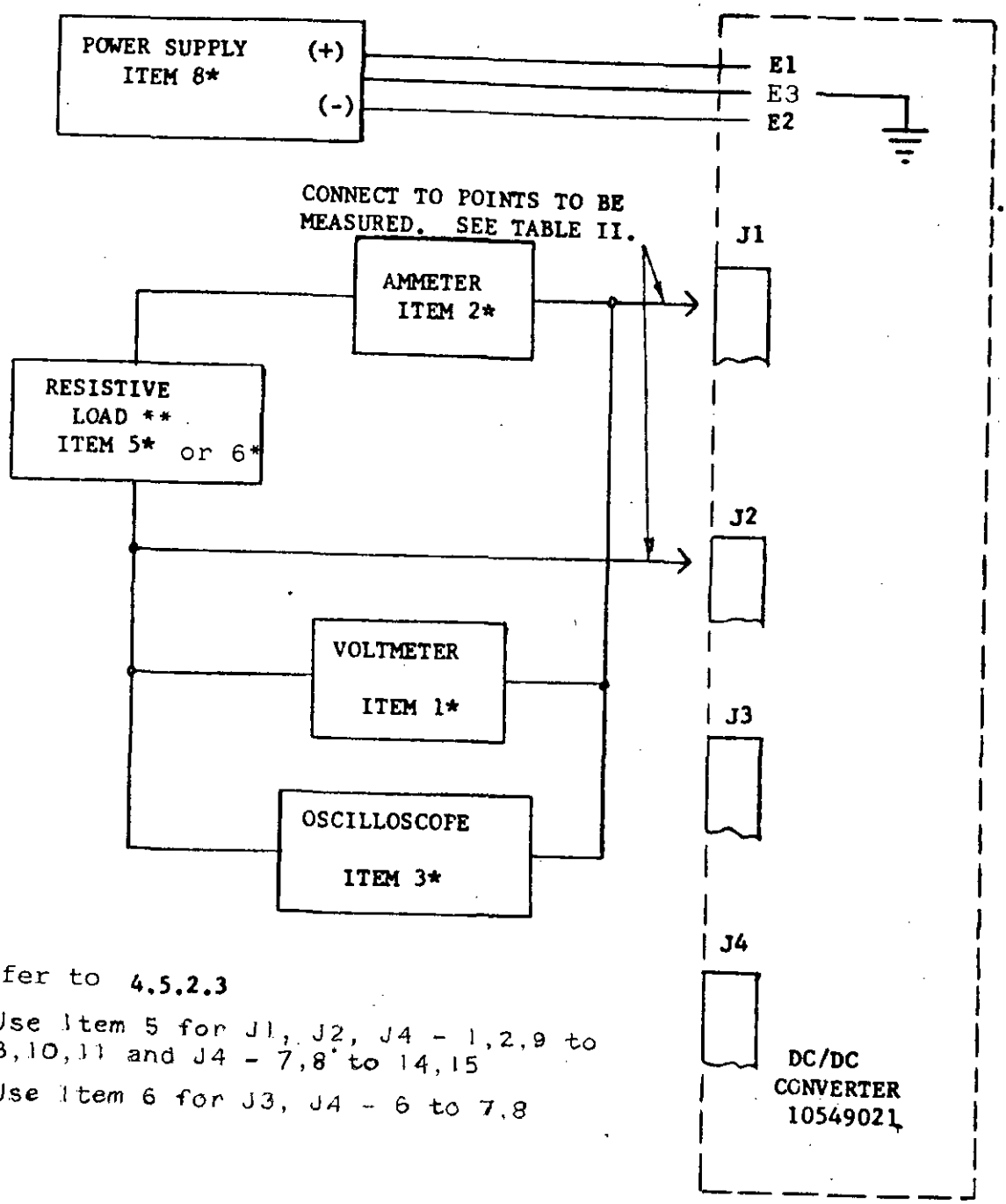


FIGURE 2. DC/DC CONVERTER TEST SETUP.

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4.6.3 Performance. The selected dc/dc converters shall be tested as follows: Apply 24^{+6}_{-2} Vdc to the unit under test using Item 8, as shown in Figure 1. Use the test equipment as shown in Figure 1 for verification of the square wave outputs as listed in Table II. Use the test equipment as shown in Figure 2 for verification of the dc outputs listed in Table II. Adjust variable resistors, Items 5 and 6, to set the output currents of Table II.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10549021. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The output voltages of the dc/dc converter are used to provide the unregulated voltages to the voltage regulators used in the 5A3 assembly, F10549013 PP-4812 AN/VPS-2, part of Radar Set AN/VPS-2. The dc/dc converter is also used in Test Set Subassembly MX8051/TPM-22, part of Test Set AN/TPM-22. In the test set it is used to check voltage regulator performance of Radar Power Supply PP-4812/VPS-2, part of Radar Set AN/VPS-2.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

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6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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